

# 1 Dynamic Programming

## 1.1 Deterministic Dynamic Programming: Review

The numbering of Assumptions, Lemmas and Theorems follows Stokey et al. (1989) (SLP thereafter).

### Sequence Problem

Find a sequence  $\{x_{t+1}\}_{t=0}^{\infty}$  such that

$$\begin{aligned} v^*(x_0) = \sup_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1}) \quad (SP) \\ \text{s.t.} \quad x_{t+1} \in \Gamma(x_t) \quad \forall t, \\ x_0 \text{ given,} \end{aligned}$$

where  $X$  is the set of possible values for the variable  $x$ ,  $F: X \times X \rightarrow \mathbb{R}$  is the return function, and  $\Gamma: X \rightarrow X$  is the correspondence describing the feasibility constraints.

### Recursive Problem: Bellman Equation

Find a function  $v: X \rightarrow \mathbb{R}$  such that

$$v(x) = \sup_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}. \quad (FE)$$

This is a functional equation; its solution is a function  $v$ , the same on both sides of (FE), that must satisfy this equality for all  $x \in X$ . If the function  $v$  (or its properties) were known, then (FE) is a two-period problem!

### Bellman's Principle of Optimality

Call any sequence  $\{x_t\}_{t=0}^{\infty}$  in  $X$  a plan. Given  $x_0 \in X$ , let  $\Pi(x_0) = \{\{x_t\}_{t=0}^{\infty} : x_{t+1} \in \Gamma(x_t), t = 0, 1, \dots\}$  be the set of plans that are feasible from  $x_0$ .

**Assumption 4.1.**  $\Gamma(x)$  is nonempty  $\forall x \in X$ .

**Assumption 4.2.**  $\forall x_0 \in X$  and  $\forall (x_0, x_1, \dots) \in \Pi(x_0)$ ,

$$\lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t F(x_t, x_{t+1})$$

exists (although it may be  $+\infty$  or  $-\infty$ ).

If Assumptions 4.1–4.2 hold, then the set of feasible plans  $\Pi(x_0)$  is nonempty for each  $x_0 \in X$  and the objective function in  $(SP)$  is well defined for every plan, so we can define the supremum,  $v^*(x_0)$ .

The function  $v^*$  is always uniquely defined (provided Assumptions 4.1–4.2 hold), whereas  $(FE)$ , for all we know, may have zero, one, or many solutions.

**Theorem 4.2.** Let  $X$ ,  $\Gamma$ ,  $F$ , and  $\beta$  satisfy Assumptions 4.1–4.2. Then the function  $v^*$  satisfies  $(FE)$ .

**Theorem 4.3.** Let  $X$ ,  $\Gamma$ ,  $F$ , and  $\beta$  satisfy Assumptions 4.1–4.2. If  $v$  is a solution to  $(FE)$  and satisfies  $\lim_{n \rightarrow \infty} \beta^n v(x_n) = 0$ ,  $\forall (x_0, x_1, \dots) \in \Pi(x_0)$ ,  $\forall x_0 \in X$ , then  $v = v^*$ .

That is,  $v^*$  is the only solution to the FE that satisfies a certain boundedness condition.

*Why is this progress?* Because it says that the two-period problem in  $(FE)$  is equivalent to the infinitely dimensional problem in  $(SP)$ . Once we have  $v$  and  $g$  (the policy function), we have the solution to  $(SP)$  for any initial condition  $x_0 \in X$ .

Next, we will apply dynamic programming techniques to study our FE.

## Bounded Dynamic Programming

**Assumption 4.3.**  $X \subseteq \mathbb{R}^l$  convex,  $\Gamma : X \rightarrow X$  is nonempty, compact-valued, continuous.

**Assumption 4.4.**  $F : Gr(\Gamma) \rightarrow \mathbb{R}$  is bounded, continuous, and  $\beta \in (0, 1)$ , where  $Gr(\Gamma) = \{(x, y) \in X \times X, y \in \Gamma(x)\}$

Define the operator  $T$  on  $C(X)$  (the space of bounded continuous functions) by

$$(Tf)(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta f(y)\}.$$

**Things you are expected to know:**

- Definition of a contraction.
- Contraction Mapping Theorem (Th. 3.2): loosely speaking, if  $T$  is a contraction,  $\exists!$  fixed point  $v = Tv$ , and iterations of  $T$  converge to  $v$ .
- Corollary to CMT:

Let  $S$  be a complete metric space, and let  $S'$  be a closed subset of  $S$ . Let  $T$  be a contraction on  $S$  and let  $v = Tv$ .

- If  $TS' \subseteq S'$ , then  $v \in S'$ .
- Moreover, if  $S'' \subseteq S'$  and  $TS' \subseteq S''$ , then  $v \in S''$ .

- Blackwell's Sufficient Conditions (Th. 3.3): monotonicity + discounting  $\implies$  contraction.
- Applying these techniques to our (FE):

**Theorem 4.6.** *Given Assumptions 4.3–4.4  $T$  maps  $C(X)$  into itself,  $T : C(X) \rightarrow C(X)$ . Moreover,  $T$  is a contraction.*

**Theorem 4.7.** *Suppose  $X$ ,  $\Gamma$ ,  $F$ , and  $\beta$  satisfy Assumptions 4.3–4.4. Assume that  $F(\cdot, y)$  is increasing,  $\Gamma$  is monotone:  $\Gamma(x) \subseteq \Gamma(x') \forall x \leq x'$ . Then the fixed point  $v$ ,  $v = Tv$ , is increasing. If  $F(\cdot, y)$  is strictly increasing, so is  $v$ .*

**Theorem 4.8.** *Suppose  $X$ ,  $\Gamma$ ,  $F$ , and  $\beta$  satisfy Assumptions 4.3–4.4. Assume  $\Gamma$  is convex:  $y \in \Gamma(x), y' \in \Gamma(x') \implies \forall x, x' \in X, \theta \in [0, 1] : \theta y + (1 - \theta)y' \in \Gamma(\theta x + (1 - \theta)x')$ . Assume  $F$  is concave in  $(x, y)$ :  $F(\theta x + (1 - \theta)x', \theta y + (1 - \theta)y') \geq \theta F(x, y) + (1 - \theta)F(x', y'), \forall (x, y), (x', y') \in Gr(\Gamma), \forall \theta \in [0, 1]$ .*

*Then the fixed point  $v$ ,  $v = Tv$ , is concave. Moreover, if  $F(\cdot, y)$  is strictly concave (inequality is strict if  $x \neq x'$  and  $\theta \in (0, 1)$ ), then  $v$  is strictly concave and  $G(x) = \{y \in \Gamma(x) : v(x) = F(x, y) + \beta v(y)\}$  is a continuous single-valued function ( $g(x)$ ).*

- **Differentiability of the Value Function**

- FOC:  $F_y(x, g(x)) + \beta v'(g(x)) = 0$
- Envelope condition:  $v'(x) = F_x(x, g(x))$

Differentiate  $v(x) = F(x, g(x)) + \beta v(g(x))$ , assuming  $g$  is differentiable

$$v'(x) = F_x(x, g(x)) + g'(x) \underbrace{[F_y(x, g(x)) + \beta v'(g(x))]}_{=0} = F_x(x, g(x)).$$

Note: this is only a simple version of the proof. The actual proof does not require that  $g$  is differentiable.

## 1.2 Introduction to Stochastic Dynamic Programming

Suppose now there is an exogenous stochastic shock. Let  $Z$  be the set of possible values for this shock. Moreover, suppose that the exogenous shock follows a Markov process, that is,  $\Pr(z_t = z | z^{t-1}) = \Pr(z_t = z | z_{t-1})$ , where  $z^t = \{z_1, \dots, z_t\}$  is a history up to period  $t$ . If  $Z$  is a finite or countably infinite set, i.e.,  $Z = \{\bar{z}_1, \bar{z}_2, \dots\}$ , then generalizing our results from deterministic dynamic programming (DDP) is completely straightforward. Denote  $\pi_{ij} = \Pr(z_{t+1} = \bar{z}_j | z_t = \bar{z}_i)$  for all  $t$ . Then in a stochastic growth model we can write our Bellman equation as

$$v(k, \bar{z}_i) = \max_{k' \in [0, \bar{z}_i f(k) + (1-\delta)k]} \left\{ U(\bar{z}_i f(k) + (1-\delta)k - k') + \beta \sum_{j=1}^{\infty} v(k', \bar{z}_j) \pi_{ij} \right\},$$

or in the general case,

$$v(x, \bar{z}_i) = \max_{y \in \Gamma(x, \bar{z}_i)} \left\{ F(x, y, \bar{z}_i) + \beta \sum_{j=1}^{\infty} v(y, \bar{z}_j) \pi_{ij} \right\}.$$

Our previous results from Deterministic Dynamic Programming (DDP) extend to this case. It is not always possible to do that for more general specifications of  $Z$ —we need to worry about issues of measurability and integrability. Chapters 7 to 9 in SLP generalize our results of DDP to stochastic environments for general specifications of  $Z$ .

Define:  $Z$ —the set of exogenous shocks,  $\mathcal{Z}$ —a  $\sigma$ -algebra on  $Z$ .

**Definition.** Let  $Z$  be a set and let  $\mathcal{Z}$  be a family of subsets on  $Z$ .  $\mathcal{Z}$  is called a  **$\sigma$ -algebra** if

a)  $\emptyset, Z \in \mathcal{Z}$

b)  $A \in \mathcal{Z} \Rightarrow A^c = Z \setminus A \in \mathcal{Z}$

c)  $A_n \in \mathcal{Z}, n = 1, 2, 3, \dots \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{Z}$

That is, a  $\sigma$ -algebra is closed under complementation and countable union. Since  $\bigcap_{n=1}^{\infty} A_n = (\bigcup_{n=1}^{\infty} A_n^c)^c$ , it is also closed under countable intersection.

When  $Z = \mathbb{R}$ , the smallest  $\sigma$ -algebra that contains all open sets is called the **Borel algebra** for  $\mathbb{R}$ . Any set in the Borel algebra is called a **Borel set**. That is,  $\mathcal{Z}$  contains all the sets of the form  $(-\infty, b)$ ,  $(a, b)$ ,  $(a, \infty)$ ,  $(-\infty, \infty)$ .

**Exercise.** Show it must also contain all closed intervals.

**Definition.** A pair  $(Z, \mathcal{Z})$ , where  $Z$  is a set and  $\mathcal{Z}$  is a  $\sigma$ -algebra of its subsets is called a **measurable space**.

**Definition.** Let  $(Z, \mathcal{Z})$  be a measurable space. A **measure** is an extended real-valued function  $\mu : \mathcal{Z} \rightarrow \overline{\mathbb{R}}$ :

a)  $\mu(\emptyset) = 0$ ,

b)  $\mu(A) \geq 0 \forall A \in \mathcal{Z}$ ,

c) if  $\{A_n\}_{n=1}^{\infty}$  is a countable, pairwise disjoint sequence of elements in  $\mathcal{Z}$ , then  $\mu(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$ .

If  $\mu(Z) = 1$ , then  $\mu$  is a **probability measure**, and  $(Z, \mathcal{Z}, \mu)$  is called a **probability space**. Then  $A \in \mathcal{Z}$  is called an **event**, and  $\mu(A)$  is called the probability of the event  $A$ .

**Definition.** Given a measurable space  $(Z, \mathcal{Z})$ , a real-valued function  $f : Z \rightarrow \mathbb{R}$  is **measurable with respect to  $\mathcal{Z}$**  (or  **$\mathcal{Z}$ -measurable**) if  $\{z \in Z : f(z) \leq a\} \in \mathcal{Z}$ , for all  $a \in \mathbb{R}$ .

If  $(Z, \mathcal{Z}, \mu)$  is a probability space and  $f$  is a  $\mathcal{Z}$ -measurable function,  $f$  is called a **random variable**.

*Example.* Living on the 56th floor of a Chicago skyscraper (good old days):

$Z = \{\text{sunshine, fog w/o rain, fog w/ rain}\},$

$\mathcal{Z} = \{\emptyset, \{\text{sunshine, fog w/o rain, fog w/ rain}\}, \{\text{sunshine}\}, \{\text{fog w/o rain, fog w/ rain}\}\},$

$$\text{utility} = f(z) = \begin{cases} 10, & \text{if } z = \text{sunshine} \\ 5, & \text{if } z = \text{fog w/o rain} \\ -5, & \text{if } z = \text{fog w/ rain} \end{cases}$$

Take  $a = 0 \Rightarrow \{z \in Z : f(z) \leq 0\} = \{\text{fog w/ rain}\} \notin \mathcal{Z}!$

To be able to write down our Bellman equation, we need a concept of a transition function.

**Definition.** Let  $(Z, \mathcal{Z})$  be a measurable space. A **transition function** is a function  $Q : Z \times \mathcal{Z} \rightarrow [0, 1]$  such that

a)  $\forall z \in Z, Q(z, \cdot)$  is a probability measure on  $(Z, \mathcal{Z})$ ,

b)  $\forall A \in \mathcal{Z}, Q(\cdot, A)$  is a  $\mathcal{Z}$ -measurable function.

Interpretation:  $Q(a, A) = \text{Pr}(z_{t+1} \in A \mid z_t = a)$

*Example.* Let  $Z = \{\bar{z}_1, \bar{z}_2, \bar{z}_3\}$ , and

$$\Pi = \{\pi_{ij}\} = \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

(rows correspond to today's states, columns to tomorrow's states, so rows sum up to one). Then, for example,  $Q(\bar{z}_2, \bar{z}_1) = 0.1$ ,  $Q(\bar{z}_2, \bar{z}_2) = 0.7$ , and  $Q(\bar{z}_2, \bar{z}_3) = 0.2$ .

If  $A = \{\bar{z}_2, \bar{z}_3\}$ ,  $Q(\bar{z}_2, A) = 0.7 + 0.2$ ,  $Q(\bar{z}_1, A) = 0.2$ , and  $Q(\bar{z}_3, A) = 0.8$ .

Our FE can be written as

$$v(x, z) = \max_{y \in \Gamma(x, z)} \left\{ F(x, y, z) + \beta \int_Z v(y, z') Q(z, dz') \right\}$$

and let  $T$  be the associated operator.

We can generalize the principle of optimality and other results. The generalization of the principle of optimality is somewhat tedious (see Section 9.1 in SLP), and the

complication is due to measurability issues that arise in the stochastic case and that have no counterpart in deterministic settings. We are going to skip it here and move on to other results.

### Bounded Returns

The following Assumptions 9.4–9.6 are the analogs of Assumption 4.3, and Assumption 9.7 is the analog of Assumption 4.4.

**Assumption 9.4.**  *$X$  is a convex Borel set in  $\mathbb{R}^l$ , with its Borel subsets  $\mathcal{X}$ .*

**Assumption 9.5.** *One of the following holds:*

- a)  *$Z$  is a countable set and  $\mathcal{Z}$  is the  $\sigma$ -algebra containing all subsets of  $Z$ ; or*
- b)  *$Z$  is a compact (Borel) set in  $\mathbb{R}^k$ , with its Borel subsets  $\mathcal{Z}$ , and the transition function  $Q$  on  $(Z, \mathcal{Z})$  has the **Feller property**, namely, the operator  $M$  defined by  $(Mf)(z) = \int f(z')Q(z, dz')$   $\forall z$ , maps the space of bounded continuous functions on  $Z$  into itself, i.e.,  $M: C(Z) \rightarrow C(Z)$ .*

**Lemma 9.5.** *Let  $(X, \mathcal{X})$ ,  $(Z, \mathcal{Z})$  and  $Q$  satisfy Assumptions 9.4–9.5.*

*Define  $S = X \times Z$ . If  $f: X \times Z \rightarrow \mathbb{R}$  is bounded and continuous, then  $\tilde{M}f$  defined by*

$$(\tilde{M}f)(y, z) = \int f(y, z')Q(z, dz') \quad \forall (y, z) \in X \times Z$$

*is also; that is,  $\tilde{M}: C(S) \rightarrow C(S)$ . If  $f$  is increasing (strictly increasing) in its first  $l$  arguments, then so is  $\tilde{M}f$ ; and if  $f$  is concave (strictly concave) jointly in its first  $l$  arguments, then so is  $\tilde{M}f$ .*

**Assumption 9.6.** *The correspondence  $\Gamma: X \times Z \rightarrow X$  is non-empty, compact-valued, and continuous.*

**Assumption 9.7.** *The function  $F: \underbrace{\text{Gr}(\Gamma)}_{\equiv A} \rightarrow \mathbb{R}$  is bounded and continuous, and  $\beta \in (0, 1)$ .*

Then we have

**Theorem 9.6.** *Let  $(X, \mathcal{X})$ ,  $(Z, \mathcal{Z})$ ,  $Q, \Gamma, F$ , and  $\beta$  satisfy Assumptions 9.4–9.7 and define the operator  $T$  on  $C(S)$  by*

$$(Tf)(x, z) = \sup_{y \in \Gamma(x, z)} \left\{ F(x, y, z) + \beta \int f(y, z')Q(z, dz') \right\}$$

Then  $T : C(S) \rightarrow C(S)$ ;  $T$  has a unique fixed point  $v$  in  $C(S)$ , and for any  $v_0 \in C(S)$ ,

$$\|T^n v_0 - v\| \leq \beta^n \|v_0 - v\|, \quad n = 1, 2, \dots$$

Similarly, we can extend our results such that  $v(\cdot, z) : X \rightarrow \mathbb{R}$  is strictly increasing and strictly concave, and  $G(\cdot, z) : X \rightarrow X$  is a continuous (single-valued) function.

**Assumption 9.8.** For each  $(y, z) \in X \times Z$ ,  $F(\cdot, y, z) : \underbrace{A_{yz}}_{\text{section of } A} \rightarrow \mathbb{R}$  is strictly increasing.

**Assumption 9.9.** For each  $z \in Z$ ,  $\Gamma(\cdot, z) : X \rightarrow X$  is increasing in the sense that  $x \leq x' \Rightarrow \Gamma(x, z) \subseteq \Gamma(x', z)$ .

**Theorem 9.7.** Let  $(X, \mathcal{X})$ ,  $(Z, \mathcal{Z})$ ,  $Q, \Gamma, F$ , and  $\beta$  satisfy Assumptions 9.4–9.9, and let  $v$  be the unique fixed point of the operator  $T$ . Then for each  $z \in Z$ ,  $v(\cdot, z) : X \rightarrow \mathbb{R}$  is strictly increasing.

**Assumption 9.10.** For each  $z \in Z$ ,  $F(\cdot, \cdot, z) : A_z \rightarrow \mathbb{R}$  satisfies

$$F[\theta(x, y) + (1 - \theta)(x', y'), z] \geq \theta F(x, y, z) + (1 - \theta)F(x', y', z)$$

for all  $\theta \in (0, 1)$  and all  $(x, y), (x', y') \in A_z$ ; and the inequality is strict if  $x \neq x'$ .

**Assumption 9.11.** For all  $z \in Z$  and all  $x, x' \in X$ ,  $y \in \Gamma(x, z)$  and  $y' \in \Gamma(x', z)$  implies

$$\theta y + (1 - \theta)y' \in \Gamma[\theta x + (1 - \theta)x', z]$$

for all  $\theta \in [0, 1]$ .

**Theorem 9.8.** Let  $(X, \mathcal{X})$ ,  $(Z, \mathcal{Z})$ ,  $Q, \Gamma, F$ , and  $\beta$  satisfy Assumptions 9.4–9.7, 9.10–9.11; let  $v$  be the unique fixed point of the operator  $T$ ; and let  $G$  be the corresponding optimal correspondence. Then for each  $z \in Z$ ,  $v(\cdot, z) : X \rightarrow \mathbb{R}$  is strictly concave and  $G(\cdot, z) : X \rightarrow X$  is a continuous (single-valued) function.

In addition, one can establish some properties of  $v(x, \cdot)$  ( $v$  as a function of  $z$ ). For example, in order to show that  $v(x, \cdot) : Z \rightarrow \mathbb{R}$  is strictly increasing, we need to impose the following assumptions (in addition to Assumptions 9.4–9.7):

**Assumption 9.13.** For each  $(x, y) \in X \times X$ ,  $F(x, y, \cdot)$  is strictly increasing.



**Assumption 9.14.** For each  $x \in X$ ,  $\Gamma(x, \cdot): Z \rightarrow X$  is increasing in the sense that  $z \leq z' \Rightarrow \Gamma(x, z) \subseteq \Gamma(x, z')$ .

**Assumption 9.15.**  $Q$  is **monotone**; that is, if  $f: Z \rightarrow \mathbb{R}$  is nondecreasing, then the function  $(Mf)(z) = \int f(z')Q(z, dz')$  is also nondecreasing.

## References

STOKEY, N., R. LUCAS, AND E. PRESCOTT (1989): *Recursive Methods in Economic Dynamics*, Cambridge, Massachusetts: Harvard University Press.